

Integració de foliacions singulars usant camins

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Outline:

- Intro to regular foliations
 - Groupoids, holonomy, paths (Haefliger, Molino, Ehresmann)
- Books

New

- Intro to singular foliations (Androulidakis, Skandalis, Debord)
- holonomy, paths for sing. foliations (A, S, D, Fernandez, Crainic, Villatoro, Zambon, G)

Regular Foliations

Infinitesimal definition

A regular foliation on a manifold M is a collection of vector fields $\mathcal{F} \subseteq \mathfrak{X}(M)$

such that for each point $p \in M$

there is some coordinates

$$(x_1, \dots, x_n) : \bigcup_{\substack{U \\ M}} \longrightarrow \mathbb{R}^n$$

with

$$\mathcal{F}|_U = \text{Span}_{C^\infty(U)} \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_k} \right)$$

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Global definition

A reg. fol. on a mfd. M is an equivalent relation $\sim$ (or a partition) s.t. $\forall p \in M \exists$ coord $U \rightarrow \mathbb{R}^n$
 $q \mapsto (x_1(q), \dots, x_n(q))$
where

$$\begin{array}{l} q_1 \sim q_2 \\ \text{in } U \end{array}$$



$$\begin{cases} x_{k+1}(q_1) = x_{k+1}(q_2) \\ \vdots \\ x_n(q_1) = x_n(q_2) \end{cases}$$

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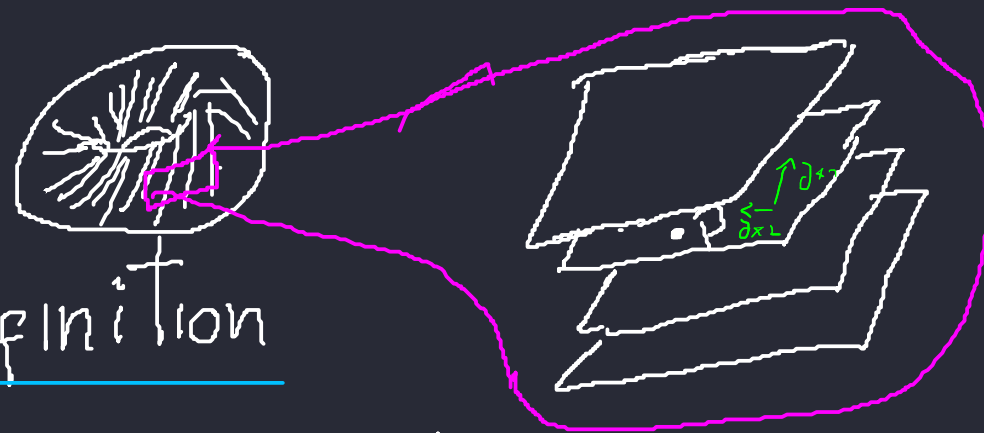
where

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Flow box theorem

The vector fields

$$Y_1, \dots, Y_k \in \mathfrak{X}(M)$$

are of the form $\frac{\partial}{\partial x_i}$ for some
coordinates $(x_1, \dots, x_n)$ near $p \in M$

iff
 $\exists$ nghd $U \subset M$ s.t.

a) $Y_1(q), \dots, Y_k(q)$ are L.I. $\forall q \in U$

b) $[Y_i, Y_j] \Big|_U = 0 \quad \forall 1 \leq j, i \leq k$

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Frobenius theorem

A regular foliation of $\dim = k$

is equivalent to a collection of vector fields $\mathcal{F} \subseteq \mathfrak{X}(M)$

s.t.

a) $\mathcal{F}(p) = \{Y(p) : Y \in \mathcal{F}\} \subset T_p M$

is a subvector space of $\dim = k$
 $\forall p \in M$

b) $[\mathcal{F}, \mathcal{F}] \subseteq \mathcal{F}$

Groupoids

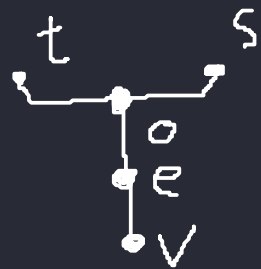
is a small category where arrows are invertible, this is

- Set of points M & set of arrows G

- Maps $s: G \rightarrow M, t: G \rightarrow M$

$\circ: G_s \times_t G \rightarrow G, e: M \rightarrow G$

$v: G \rightarrow G$



s.t.

a) $\circ$ is associative

b) for $y \xleftarrow{\varphi \in G} x \in M$ $e(y) \circ \varphi = \varphi \circ e(x) = \varphi$

c) $v(\varphi) \circ \varphi = e(x), \varphi \circ v(\varphi) = e(y)$

Groupoids

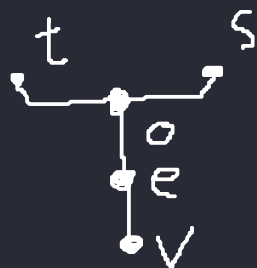
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Example

Let M be a set with an eq. rel. $\sim$

Take $G = \text{Graph}(\sim) = \{(y, x) \in M \times M : y \sim x\}$

$s(y, x) = x, t(y, x) = y$

$(z, y) \circ (y, x) = (z, x)$ (transitivity)

$e(x) = (x, x)$ (Reflexive)

$v(y, x) = (x, y)$ (Symmetric)

Lie Groupoids

in mfd

is a small category where arrows are invertible, this is

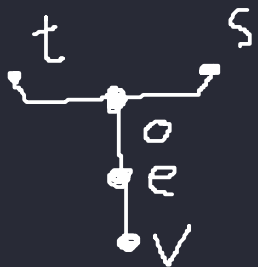
• mfd of points M & mfd of arrows G

SMOOTH

• Maps $s: G \rightarrow M$, $t: G \rightarrow M$ ← Surjective Submersion

$\circ: G_s \times_t G \rightarrow G$, $e: M \rightarrow G$

$v: G \rightarrow G$



s.t.

a) $\circ$ is associative

b) for $y \xleftarrow{\varphi \in G} x \in M$ $e(y) \circ \varphi = \varphi \circ e(x) = \varphi$

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Example

smooth

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$$t(y, x) = y$$

$$(z, y) \circ (y, x) = (z, x) \quad (\text{Transitivity})$$

$$e(x) = (x, x) \quad (\text{Reflexive})$$

$$v(y, x) = (x, y) \quad (\text{Symmetric})$$

Example (Foliation local)

In $M = \mathbb{R}^k \times \mathbb{R}^{n-k}$ as points

Take $\mathcal{G} = \{\text{paths constant in } \mathbb{R}^{n-k}\} / \text{Hom}$

$$s(\gamma) = \gamma(0)$$

$$t(\gamma) = \gamma(1)$$

$\circ$ concatenation

$e(x)$ constant path on x

$V(\gamma)$ inverse parametrization

there are charts $\mathbb{R}^k \times \mathbb{R}^k \times \mathbb{R}^{n-k} \xrightarrow{\sim} \mathcal{G}$ so it is smooth
charts $(x, y, c) \mapsto (xt + (1-t)y, c)$

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Example (General)

For a mfd M with reg. fol $\mathcal{F}$ of dim k

Take $\mathcal{G} = \{\text{paths in a class}\} / \text{Hom}$ of the class

the same $s, t, \circ, e, v$ as before
is a Lie Groupoid and

it looks locally as

$$\mathbb{R}^k \times \mathbb{R}^k \times \mathbb{R}^{n-k}$$

Singular Foliations on a mfd $M(A.S.)$

is a collection of v.f. $\mathcal{F} \subseteq \mathcal{X}_c(M)$

s.t.

a) $[\mathcal{F}, \mathcal{F}] \subseteq \mathcal{F}$ (closed under Lie brackets)

b) $\forall p \in M \exists$ nghd $U \subseteq M$ & k_p
with $\gamma_1, \dots, \gamma_{k_p} \in \mathcal{X}(U)$

s.t. $\mathcal{F}|_U = \text{Span}_{C_c^\infty(U)}(\gamma_1, \dots, \gamma_{k_p})$

Thm any sing. fol. gives an eq. rel $\sim_{\mathcal{F}}$
and each class $L_x = \{ \gamma : \gamma \sim_{\mathcal{F}} x \}$
is a submfd called a leaf

Results

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Results

• There is a def for paths and for homotopy in $\mathcal{F}$ such that:

Paths/Homotopy is an (almost Lie) groupoid (G_V)

• the almost smooth str. comes from a quotient of the mfd $\mathbb{R}^n \times U$ for U open in M (AS) & (G_V)

• Paths/Homotopy on a leaf is smooth(Lie) (AZ)

• Paths/Homotopy is a Lie groupoid

$\Rightarrow \mathcal{F}$ is almost regular (AZ)

Smooth maps to a sing. fol. in a mfd M

a map $Y: V_{\mathbb{R}^n} \rightarrow \mathcal{F}$ is smooth if
 $\forall p \in M \exists$ nhd $U \in M$ with

$$Y(v)(q) = \sum_{i \in I} f_i(t, q) Y_i(q)$$

$\forall q \in U, I$ finite, $Y_i \in \mathcal{F}$ & $f_i \in C^\infty(V \times M)$

Paths to $\mathcal{F}$ is a collection of a

[smooth map $Y: [0, 1] \rightarrow \mathcal{F}$

&

[a curve $\gamma: [0, 1] \rightarrow M$

[s.t $\dot{\gamma}(t) = Y(t, \gamma(t))$

Smooth maps to a sing. fol. in a mfd M

a map $Y: V \xrightarrow{\sim \mathbb{R}^n} F$ is smooth if
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$\forall q \in U, I$ finite, $Y_i \in F$ & $f_i \in C^\infty(V \times M)$

Paths to F is a collection of a

[smooth map $Y: [0, 1] \rightarrow F$

&
 a curve $\gamma: [0, 1] \rightarrow M$

s.t. $\dot{\gamma}(t) = Y(t, \gamma(t))$

A F -homotopy between paths in F

(Y_0, γ_0) & (Y_1, γ_1) is a smooth map

$$Y: [0, 1]^2 \rightarrow F$$

$$\gamma: [0, 1]^2 \rightarrow M \quad \gamma(s, 0) = \gamma_0(s) = \gamma_1(s)$$

s.t.

- $Y(0, t, q) = Y_0(t, q)$ & $Y(1, t, q) = Y_1(t, q)$
 $\gamma(0, t) = \gamma_0(t)$ & $\gamma(1, t) = \gamma_1(t)$

- $Y(s, t, q) = \frac{d}{dt} \gamma(t, q)$

- the vector field $W(s, t, q) = \frac{d}{ds} \Phi^{t, s}_Y$

satisfy $W(s, 1, \gamma(s, 1)) \in T_{\gamma(s, 1)} F$
 $\forall s \in [0, 1]$